DataClaw

AS-BUILT ARCHITECTURE

A local-first data science agent platform with a plugin architecture and OpenClaw integration.

DataClaw is a two-part app: a Python backend with a plugin-based agent pipeline (FastAPI + LangGraph) and a React frontend, bridged to the OpenClaw agent runtime via channel and tool plugins. This page documents what was actually built.

ARCHITECTURE STACK

```
DataClaw =
  FastAPI + LangGraph agent pipeline
  + 7 plugins (workspace, data, notebooks, plans, projects, browser, openclaw)
  + 8 swappable providers (agent, llm, compaction, system prompt, memory, skill, tools, sub-agent)
  + OpenClaw gateway (agent runtime, model routing, tool orchestration)
  + DuckDB data plane + MLflow experiment tracking
  + React UI (AG-UI protocol, Ant Design)
```

7	8	34	9	2
BACKEND PLUGINS	PROVIDER SLOTS	AGENT TOOLS	HOOK POINTS	OPENCLAW PLUGINS

01 CORE ARCHITECTURE

DataClaw is a plugin-based agent platform with OpenClaw as its recommended runtime.

The system is split into two layers: DataClaw owns the product semantics (data access, notebooks, plans, projects, workspace governance) through a Python plugin system with swappable providers. OpenClaw provides the agent runtime (model routing, multi-provider support, memory, tool orchestration) and is bridged via two TypeScript plugins.

OPENCLAW AGENT RUNTIME — External agent runtime (recommended backend)

- **Multi-model routing** across Anthropic, OpenAI, Google, and local models.
- **Gateway** with WebSocket control plane and plugin system.
- **dataclaw-frontend** channel plugin routes messages between UIs.
- **dataclaw-tools** tool plugin exposes DataClaw's 34 tools to the agent.
- **Memory, skills, and identity** managed by OpenClaw's own systems.
- **Bundled Docker deployment** runs both in a single container.

DATACLAW PRODUCT LAYER — What DataClaw owns

- **Plugin system** with entry-point discovery, dependency ordering, and provider injection.
- **Agent pipeline** (LangGraph) with 9 hook points for plugin intervention.
- **Data plane** with DuckDB, read-only safe SQL, dataset registry, profiling, and column analysis.
- **Notebook management** with isolated Jupyter kernels per project.
- **Plan proposals** with approval workflows and MLflow experiment tracking.
- **Project workspaces** with scoped sessions, file management, and dataset access control.
- **React UI** with chat, data browser, config, skills, and project management.

Design rule: DataClaw owns product semantics (connector policy, artifact identity, plan approval, workspace governance). OpenClaw owns agent capability (model routing, tool dispatch, session management). The provider layer abstracts LLM backends so routers and tools never call provider APIs directly.

02 PLUGIN EXPLORER

Click any component to see what it provides.

Filter the view to isolate DataClaw's backend plugins, the OpenClaw bridge layer, or the core infrastructure.

Show all | Backend plugins | OpenClaw bridge | Infrastructure

dataclaw-workspace — PLUGIN
File I/O and shell execution.
6 tools: list files, read/write/update files, shell exec, display images. Hooks inject project directory context.
ws_list_files · ws_exec · ws_read_file

dataclaw-data — PLUGIN
Dataset registry and DuckDB data plane.
6 tools: list datasets, preview, profile, describe column, query (read-only SQL), get docs. Routes for dataset CRUD and upload.
DuckDB · safe SQL · profiling · multi-source

dataclaw-notebooks — PLUGIN
Jupyter notebook management with isolated kernels.
13 tools: open/close/list/read notebooks, insert/edit/move/delete cells, execute code, display outputs.
Jupyter kernels · per-project venvs · nbformat

dataclaw-plans — PLUGIN
Plan proposals and MLflow experiment tracking.
5 tools: propose/update/list/get plans, query MLflow runs. Routes for plan decisions and feedback.
plan approval · MLflow · experiment runs

dataclaw-projects — PLUGIN
Project management with scoped sessions and subagents.
2 tools: list/delegate subagents. Routes for project CRUD and data source assignment.
project scoping · dataset access control · subagents

dataclaw-browser — PLUGIN
AI-driven browser automation.
1 tool: browser_use for web browsing, content extraction, and task completion.
Playwright · web scraping · feature-flagged

dataclaw-openclaw — PLUGIN
OpenClaw agent bridge and installation management.
Replaces the default agent provider with OpenClaw delegation. Routes for install, plugin management, token fetch.
agent provider swap · install UI · hot reload

dataclaw-frontend (OpenClaw) — BRIDGE
Channel plugin routing messages between DataClaw UI and OpenClaw agents.
HTTP routes for message dispatch and SSE streaming. Persists responses back to DataClaw sessions.
TypeScript · channel plugin · SSE

dataclaw-tools (OpenClaw) — BRIDGE
Tool plugin exposing DataClaw's Python tools to OpenClaw agents.
Dynamic tool discovery from GET /api/tools. Prefixes tools with dataclaw_ and injects session context.
TypeScript · tool proxy · 34 tools

Agent Pipeline (LangGraph) — CORE
13-node state machine orchestrating the agent turn loop.
Flows through: userQuery → compaction → systemPrompt → memory → skill → toolAvailability → agent → [tool loop].
LangGraph · 9 hook points · streaming

Provider Registry — CORE
8 swappable provider slots for dependency injection.
Slots: agent, llm, compaction, system_prompt, memory, skill, tool_availability, sub_agent. Plugins can replace any.
DI container · hot-reloadable · protocol-based

AG-UI Protocol — CORE
Standardized agent-to-UI communication via Server-Sent Events.
Background task execution with reconnectable event streams. Supports multi-tab, replay, and cancellation.
SSE streaming · run tracker · reconnect

03 ARCHITECTURE MAP

How the layers connect from UI to agent runtime.

Click any node in the map. The sidebar updates to explain what each component does and how it fits into the system.

Selected component
Click any node to see details about its role, capabilities, and connections.

FRONTEND
React UI
React SPA with Ant Design providing chat, data browser, notebooks, projects, skills, and configuration pages.
- AG-UI protocol over SSE for streaming.
- Reconnectable event streams with multi-tab support.
- Embedded terminal for OpenClaw CLI operations.

FRONTEND + API
- React UI — Chat, Data, Projects, Config
- FastAPI — REST API + SSE streaming
- Agent Pipeline — LangGraph 13-node state machine
- Provider Registry — 8 swappable slots

PLUGIN HOOKS ↔ TOOL CALLS

DATACLAW PLUGINS
- dataclaw-data — DuckDB + dataset registry
- dataclaw-notebooks — Jupyter + isolated kernels
- dataclaw-plans — Plans + MLflow tracking
- dataclaw-workspace — Files + shell execution
- dataclaw-projects — Projects + scoping

HTTP BRIDGE ↔ TOOL PROXY

OPENCLAW RUNTIME
- OpenClaw Gateway — Agent runtime + model routing
- dataclaw-frontend — Channel plugin (messages)
- dataclaw-tools — Tool plugin (34 tools)

04 AGENT PIPELINE

The LangGraph pipeline every message flows through.

Use the stepper to move through the agent pipeline. Each stage has hook points where plugins can inject context, filter tools, or intercept messages. The pipeline loops back on tool calls until the agent produces a final response.

Analytics Harness sequence
Step 1 of 9

Previous | Next

User query → Compaction → System prompt → Memory → Skills → Tool availability → Call agent → Tool execution → Response

User query
The user message enters the pipeline. The userQueryHook fires first, allowing plugins to intercept or modify the query before processing begins.
- Message added to AgentState with session/project context.
- userQueryHook fires (plugins can modify or reject).
- Turn counter incremented.

05 UI OVERVIEW

The DataClaw interface as built.

The React frontend provides chat with plan approval, a dataset registry, project management with scoped sessions, skill authoring, and configuration with OpenClaw integration. All connected via the AG-UI protocol.

DataClaw — OpenClaw connected · 7 plugins loaded · 34 tools available

Chat | Data | Projects | Skills | Config

SESSIONS
- Churn analysis
- Revenue forecast Q3
- Data quality audit

ACTIVE PLANS
- Profile customer_events
- Build retention model

LOADED PLUGINS
- dataclaw-workspace
- dataclaw-data
- dataclaw-notebooks
- dataclaw-plans

TOOLS AVAILABLE: 34 — Workspace (6), data (6), notebooks (13), plans (5), projects (2), browser (1), images (1).

SESSIONS: 3 — Multi-session chat with project scoping, dataset filtering, and plan sidebar.

PLANS PENDING: 1 — Agent-proposed plans require user approval before execution begins.

Current plan: Churn analysis
1. **Profile customer_events dataset** — data_profile_dataset · DuckDB — done
2. **Query churn-risk segments** — data_query_data · safe SQL — done
3. **Build retention model notebook** — open_notebook · execute_cell — running
4. **Log experiment to MLflow** — query_mlflow_runs — queued

Chat features
- Streaming responses via AG-UI SSE.
- Tool call visualization with collapsible details.
- Plans sidebar with step tracking and approval.
- File preview and MLflow experiment modals.

06Responsibility Matrix

What DataClaw owns vs what OpenClaw provides.

DataClaw owns product semantics. OpenClaw owns agent capability. This table shows who is responsible for what.

Capability	Owner	Implementation	Key details
Agent runtime	OpenClaw	Gateway + model routing	Multi-provider (Anthropic, OpenAI, Google, local), memory, identity, skills
Data access	DataClaw	dataclaw-data plugin	DuckDB engine, read-only safe SQL, dataset registry, profiling, multi-source connectors
Notebook execution	DataClaw	dataclaw-notebooks plugin	Jupyter kernels with per-project isolated venvs, 13 cell management tools
Plan approval	DataClaw	dataclaw-plans plugin	Propose/approve workflow, status tracking, user feedback capture
Experiment tracking	DataClaw	MLflow via dataclaw-plans	Auto-created experiments per session, parameters/metrics/artifacts/datasets
Project scoping	DataClaw	dataclaw-projects plugin	Scoped sessions, dataset access control, file workspaces, subagent delegation
File I/O + shell	DataClaw	dataclaw-workspace plugin	6 tools with configurable limits, project directory injection via hooks
Message bridge	Shared	dataclaw-frontend + dataclaw-openclaw	Channel plugin (TS) routes messages; bridge plugin (Python) swaps agent provider
Tool bridge	Shared	dataclaw-tools plugin (TS)	Dynamic discovery of 34 tools, session context injection, dataclaw_ prefix

07 STATUS

What's built, what's in progress, and what's next.

DataClaw's core platform is functional. The plugin system, agent pipeline, data plane, notebooks, and OpenClaw integration are all working. Here's the current state.

COMPLETE — **Core platform**
- Plugin system with entry-point discovery and DI
- LangGraph agent pipeline with 9 hooks
- 8 swappable provider slots
- FastAPI with AG-UI SSE streaming
- React UI (chat, data, projects, config, skills)

COMPLETE — **Data + notebooks**
- DuckDB data plane with safe SQL
- Dataset registry with multi-source connectors
- Profiling and column analysis tools
- Jupyter notebooks with isolated kernels
- Per-project Python venvs via uv

COMPLETE — **OpenClaw integration**
- dataclaw-frontend channel plugin
- dataclaw-tools tool bridge (34 tools)
- Bundled Docker deployment
- Non-interactive bootstrap and onboarding
- Config page with install and model auth

COMPLETE — **Plans + experiments**
- Plan proposal and approval workflow
- MLflow experiment tracking per session
- Parameter, metric, and artifact logging
- Project-level experiment browsing
- Subagent definitions and delegation

NEXT — **Hardening + expansion**
- Richer connector support (Snowflake, BigQuery)
- Analysis bundle export and validation
- Dashboard/View generation
- Official DS skill library
- Authentication and multi-user support

08 KEY TAKEAWAYS

The shortest useful summary of DataClaw's architecture.

Plugin architecture, not a monolith.
7 backend plugins discovered via entry points, dependency-sorted, with DI. Each registers tools, hooks, routes, and optionally swaps providers. Adding new capabilities means writing a plugin, not modifying the core.

DataClaw owns product semantics. OpenClaw owns agent capability.
Connector policy, artifact identity, plan approval, workspace governance, and dataset access control are DataClaw's responsibility. Model routing, tool dispatch, memory, and session management are OpenClaw's. The provider layer abstracts the boundary.

Safe SQL by default, hooks everywhere.
Dataset queries are read-only DuckDB with single-statement validation and result caps. 9 hook points let plugins inject project context, filter dataset access, and intercept tool calls without touching the pipeline core.

Architecture summary derived from the DataClaw codebase.